

Compound Weather Disruption and Passenger Stranding in a European Rail-Air Network: An Agent-Based Simulation

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Abstract. This paper simulates how passengers redistribute across a 53-city European rail-air network when compound weather events simultaneously disrupt multiple infrastructure nodes. Three stylised stress-test scenarios exercise a gravity-calibrated agent-based model (ABM) incorporating a mean-field Bass-style awareness process. Results show that stranding is determined primarily by network topology: direct node closures produce hard stranding (complete topological isolation: no path survives in the remaining network) that awareness of disruption cannot overcome, while peripheral cities with limited multimodal alternatives carry the highest stranding risk. Stranding geography is compared against Storm Ciaran (November 2023) as an external plausibility check, suggesting that network structure helps explain observed disruption severity.

Keywords: agent-based modelling; multimodal transport disruption; network vulnerability

1 Introduction

European long-distance travel relies on two modes that look independent on a map and behave as one in operation: rail and air. When a flight is cancelled at Heathrow, displaced passengers divert to Eurostar; when heat buckles track near Paris, demand spills back to air. The two modes share a demand pool, and that pool is the channel through which one mode’s disruption propagates into the other. Compound weather events (episodes in which a single climatic driver simultaneously degrades multiple infrastructure systems) are projected to intensify under climate change (IPCC, 2023; Zscheischler et al., 2020), making the assumption that travellers can always substitute across modes increasingly untenable. Storm Eunice (February 2022) illustrates this: Nederlandse Spoorwegen (NS), the Dutch national railway, suspended all rail services across the Netherlands while Amsterdam Schiphol cancelled over 200 flights simultaneously.

Buldyrev et al. (2010) formalise this as an interdependency regime in which simultaneous dual-network failure is qualitatively distinct from sequential failure of either network alone.

This paper asks how passengers redistribute across a multimodal European transport network when extreme weather simultaneously disrupts rail and air, and whether network peripherality amplifies stranding beyond the directly-disrupted cities. Demand-weighted city-level estimates for this compound regime are currently unavailable, yet directly relevant to operators, supranational coordinators (ERA, EUROCONTROL), and emergency planners. This paper fills that gap, extending Ippolito and Cats (2024)’s structural framework with passenger agents, behavioural routing, and information dynamics.

2 Background

Network vulnerability. Transport network vulnerability has been formalised since Berdica’s (2002) foundational definition: susceptibility to incidents that reduce network serviceability, measurable through accessibility degradation and redundancy loss. Subsequent work operationalised this at European scale. Ippolito and Cats (2024) analyse a 124-city European rail-air network under targeted node removal, finding that rail loses 71% of performance under worst-case attack while air loses only 10%, a structural asymmetry driven by rail’s lower degree and path

redundancy. Mattsson and Jenelius (2015) argue that this purely topological tradition understates vulnerability in demand-rich networks, proposing a system-based framework in which disruption impact is weighted by the volume of passenger flow affected. Reggiani et al. (2015), drawing on Holling (1973), distinguish *engineering resilience* (structural path availability) from *ecological resilience* (absorption while remaining functional), a contrast applied in the Results section. The binding limitation of this structural tradition is that passengers are implicit: route choices, information states, and demand distributions do not enter the analysis.

Compound weather disruption. Zscheischler et al. (2020) define compound events as combinations of drivers or hazards that individually may be manageable but jointly produce amplified societal impact. Compound transport disruptions are particularly severe because simultaneous failure of substitutable modes removes the traveller’s fall-back option, creating the interdependency regime formalised by Buldyrev et al. (2010). Under climate change, the frequency and intensity of such events are projected to increase (IPCC, 2023), making their transport consequences an urgent research priority. Despite this, the behavioural passenger response to compound disruption (which cities accumulate stranded travellers and whether peripheral cities suffer disproportionately) remains largely uncharacterised.

Agent-based modelling and information diffusion in transport. Agent-based modelling (ABM) provides a natural complement to structural vulnerability analysis: by populating the network with heterogeneous agents whose local decisions aggregate into system-level outcomes, ABM captures the demand and behavioural dimensions that graph-theoretic approaches leave implicit (Epstein and Axtell, 1996). Leng and Corman (2020) demonstrate in a Zürich public-transport disruption ABM that real-time information materially improves passenger rerouting outcomes; Cats and Jenelius (2014) formalise disruption information as a vulnerability-mitigation lever in public transport. Both studies operate at urban scale with relatively complete network coverage. Information dynamics are modelled here using the Bass (1969) diffusion framework, in which an external broadcast channel (innovation coefficient p) and peer-to-peer contact (imitation coefficient q) jointly govern awareness spread through an affected population. D’Arienzo et al. (2020) use Bass-style diffusion for emergency-warning messages, while O’Shea et al. (2020) model direct and indirect flood-warning effects in an agent-based crisis setting. Model structure follows the ODD protocol (Grimm et al., 2020) and calibration is inspired by pattern-oriented modelling (Grimm et al., 2005).

3 Model

The model extends the structural network framework of Ippolito and Cats (2024) by adding passenger agents, demand weighting, and a mean-field awareness layer; all implementation is original to this study.

3.1 Network

The transport network is a directed multigraph over 53 European cities, selected by population threshold of 250,000,¹ shown in Figure 1. Each city is a single node regardless of the number of airports or stations it contains, following Ippolito and Cats (2024). The simplified undirected projection in Figure 1 contains 717 mode-specific edges. The directed multigraph used in simulation contains 1,434 reciprocal directed arcs plus two additional asymmetric one-way arcs, for 1,436 directed arcs in total; each arc is weighted by travel time and used for shortest-path routing (data sources and travel-time derivation in Section 4).

¹Eurostat Urban Audit, ec.europa.eu/eurostat/web/cities, accessed April 2026.

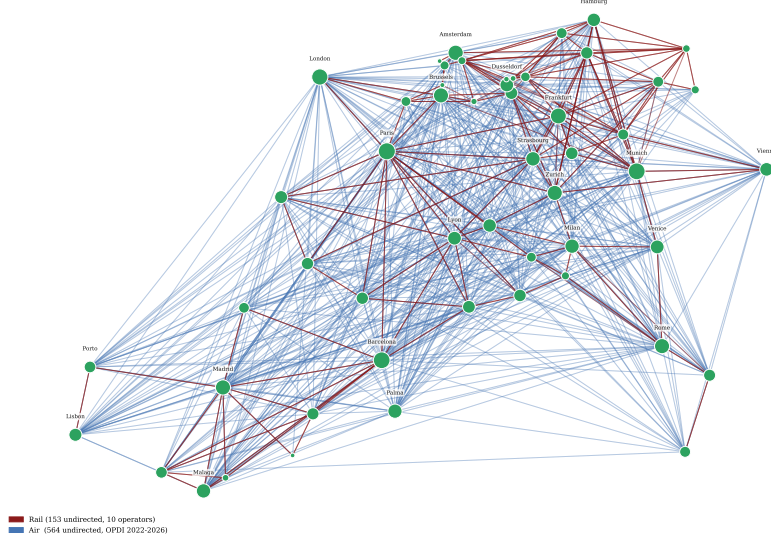


Figure 1: 53-city European multimodal transport network. Node size proportional to degree.

3.2 Agents and Demand

Each agent carries: origin, destination, current node, planned path, mode preference, information state, and status (*travelling*, *stranded*, *arrived*, or *rescheduled*). Each agent’s origin-destination (OD) pair is drawn from 902 directed city pairs proportionally to gravity-weighted traffic volume; mode is assigned by route-specific modal split (Section 4, Equation 6).

The *stranding index* for city c is:

$$\sigma_c = \frac{\bar{N}_c^{\text{strand}}}{N_c^{\text{OD}}}, \quad (1)$$

where $\bar{N}_c^{\text{strand}}$ is mean stranded agents whose origin or destination is city c (regardless of where stranding occurs), averaged across Monte Carlo runs, and N_c^{OD} is total agents with city c as origin or destination. This exposure-normalised metric follows Jenelius et al. (2006): dividing by traffic volume enables fair comparison across cities of different sizes. Because stranding is topological isolation rather than capacity saturation, the absolute agent count does not affect which cities strand; the model needs only enough agents for the gravity-weighted OD distribution to be well-sampled.

The simulation generates 100 new agent journeys per hour ($\lambda = 100 \text{ h}^{-1}$), a scale chosen so that per-city stranding counts are interpretable without excessive runtime. Since the average journey across all active OD pairs takes 2.09 hours (gravity-weighted; Section 4), at any moment under normal conditions roughly $100 \times 2.09 \approx 209$ agents are in transit (Little’s Law, Little 1961, applied to the simulation steady state). Five Monte Carlo runs are performed per scenario; results report mean $\pm$ SD. Because hard stranding is defined topologically, Monte Carlo variation affects realised agent counts but not the existence of a surviving path; five runs are therefore used for count averaging, while sensitivity tests (Section 5) confirm that city rankings remain stable across a tenfold range of agent population.

3.3 Dynamics and Disruption Scenarios

Each of 72 one-hour steps (a three-day simulation horizon), agents follow the shortest time-path through currently available (non-disrupted, time-active) edges. Air and day-rail services operate 05:00–22:59; night-rail services operate 20:00–08:59; edges outside their service window are

temporarily unavailable, producing overnight dips in movement rates visible in the dynamics plots. At each step, three decision stages apply: (1) If the agent is *informed* and its planned path contains a disrupted edge, it either opts not to travel (exits the network) with a calibrated probability, or reroutes proactively around the blockage. (2) If the agent is *uninformed* and attempts to board a blocked edge, it discovers the disruption at that point (experiential contact), becomes informed, and attempts to find an alternative path from its current position. (3) If no path from its current node to its destination exists through the remaining network (topological isolation, not delay), the agent becomes *stranded*.

The key difference is positional: uninformed agents discover the disruption mid-journey, from a potentially worse network position (Figure 2).

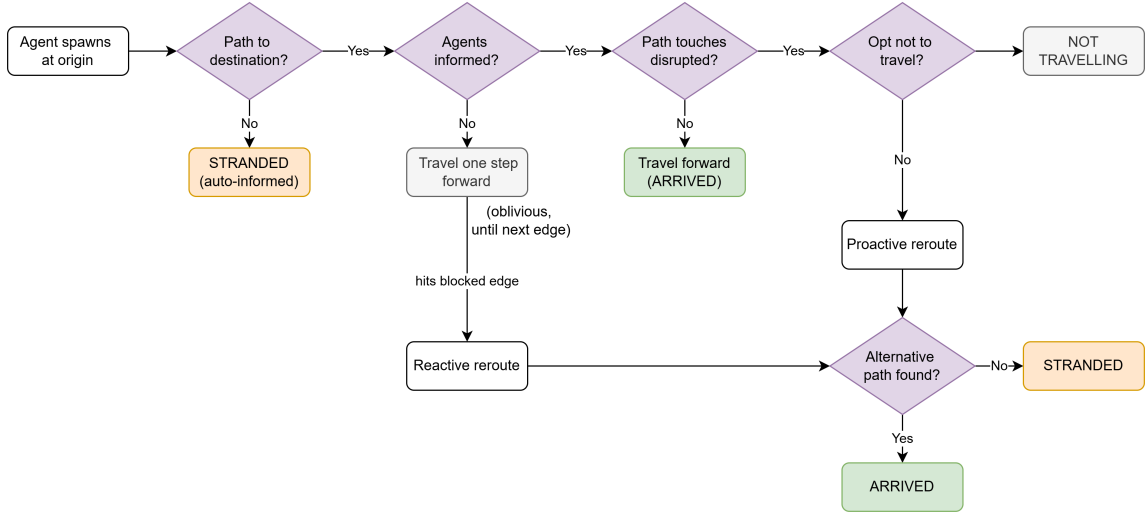


Figure 2: Agent decision logic per simulation step.

Three stylised stress-test scenarios are evaluated (Table 1), each targeting a different network mechanism: hub closure, corridor disruption, and peripheral isolation. Each scenario is static: following a 24-step warm-up of undisrupted travel, disruptions are applied at hour 24 and persist for the remaining 48 steps with no recovery, isolating the network-topology mechanism from temporal dynamics.

Table 1: Disruption scenario specifications.

Scenario	Closed nodes	Closed corridors	Primary mechanism
A: Northern Storm	London, Amsterdam, Hamburg	–	Hub closure
B: Alpine Freeze	Zurich	28 Alpine rail edges	Corridor disruption
C: Atlantic Cascade*	Lisbon, Porto	Iberian rail + Paris CDG air	Peripheral isolation

* CDG air connections to Iberia are disrupted alongside Iberian rail, instantiating a spatially compounding event in the typology of Zscheischler et al. (2020).

3.4 Information Layer

Following Leng and Corman (2020) and Cats and Jenelius (2014), an awareness layer tracks whether each agent is informed of the disruption; the approach builds on information-contagion models for transit disruptions (Hua and Ong, 2018). Awareness spreads according to a Bass-type diffusion equation:

$$\frac{\Delta I}{\Delta t} = \underbrace{p_{\text{int}}(t) \cdot S}_{\text{broadcast}} + \underbrace{q_{\text{contact}} \cdot \frac{I \cdot S}{N}}_{\text{peer contact}}, \quad (2)$$

where I is the number of informed agents, $S = N - I$ the number of uninformed, N the total active agents, p_{int} is the broadcast rate (mass media and app notifications), q_{contact} the peer-contact rate, and $I(0) = p_{\text{init}} \cdot N$ is the initial informed share (agents already aware of the disruption at onset). The contact term $q_{\text{contact}} \cdot IS/N$ is mean-field (modelling phone and social-media contact), not diffusion over a co-location network. Two further channels operate outside Equation 2: an agent reaching a blocked edge discovers the disruption immediately (*experiential*), and a topologically stranded agent is automatically informed (*forced*). All four channels are decomposed in Table 4.

4 Data and Calibration

Calibration is inspired by the pattern-oriented modelling (POM) approach of Grimm et al. (2005): summary statistics from observed behaviour are matched across multiple patterns rather than a single aggregate.

4.1 Network Data

Rail edges are constructed from General Transit Feed Specification (GTFS) timetables of national operators of relevant countries.² A frequency cap of 20 trains/day restricts edges to intercity services; commuter routes are excluded. Routing uses Dijkstra on the time-weighted directed graph; service windows are 05:00–22:59 for air and day rail, 20:00–08:59 for night rail.

Air edges are derived from Open Performance Data Initiative (OPDI) flight-list data³ (January 2022–March 2026). Travel time equals haversine distance divided by 800 km/h (ESPON, 2013) plus a 45 min boarding overhead as a conservative modelling assumption informed by airport-process guidance (IATA, 2019).

4.2 Gravity Demand Calibration

OD volumes are estimated using structural gravity (Anderson and van Wincoop, 2003) with Poisson Pseudo-Maximum-Likelihood (PPML; Santos Silva and Tenreyro 2006), recently applied to German domestic and neighbouring-country rail-air substitution by Oesingmann and Ennen (2025). Separate models are fitted for air and rail (Table 2).

The **air model** includes origin and destination city fixed effects:

$$V_{ij}^{\text{air}} = \exp(\alpha_i + \beta_j - \gamma_{\text{air}} \log d_{ij}). \quad (3)$$

The **rail model** uses log travel time with symmetric population and a same-country dummy (city fixed effects are excluded):⁴

$$V_{ij}^{\text{rail}} = \exp(\theta \log(P_i P_j) + \phi |\log P_i - \log P_j| - \gamma_{\text{rail}} \log t_{ij} + \delta \mathbf{1}[\text{same country}]). \quad (4)$$

The GTFS proxy measures supply frequency rather than observed OD demand, a limitation inherent to network-scale European rail modelling at this geographic scope; the level R^2 reflects structural noise in the proxy rather than a failure of the gravity specification. The ABM uses demand weights for stochastic OD sampling, not for point prediction of passenger counts: rank order of pairs determines which corridors attract most agents, and the Spearman correlation of

²GTFS feeds accessed April 2026: DB; SNCF; ÖBB; Renfe; SBB; Trenitalia; NS/OVapi; SNCB; CP; Eurostar GTFS, accessed 30 April 2026.

³OPDI flight-list data, accessed 30 April 2026.

⁴The rail OD set contains 134 pairs; including city fixed effects (FE) would require ≈ 51 FE parameters, producing catastrophic overfit confirmed in preliminary runs.

0.611 (Table 2) confirms this ordering is preserved in the test set, supporting its use for coarse demand-weighted OD sampling. Route-specific modal split:

$$s_{ij}^{\text{air}} = \frac{V_{ij}^{\text{air}}}{V_{ij}^{\text{air}} + V_{ij}^{\text{rail}}}. \quad (5)$$

Because V_{ij}^{rail} is derived from GTFS service frequency (a supply proxy) rather than observed OD passenger counts, this ratio should be interpreted as a calibrated supply-weighted routing prior rather than an empirically observed modal share. The OD sampling weight used by the ABM is:

$$w_{ij} = s_{ij}^{\text{air}} \cdot V_{ij}^{\text{air}} + (1 - s_{ij}^{\text{air}}) \cdot V_{ij}^{\text{rail}}. \quad (6)$$

Table 2: Gravity calibration results: v4 dual-mode model.

Mode	Estimator	Specification	$\hat{\gamma}$	R^2 (test)	Spearman ρ
Air	PPML + city FE	log-distance	0.189	0.610	0.793
Rail	PPML, no FE	log-travel-time + sym-pop + same-country	0.422	0.333	0.611

4.3 Behavioural Prior Calibration and Event-Level Plausibility Check

Information diffusion. Because no transport-specific estimates exist for hourly information acquisition during a pan-European multimodal disruption, p_{int} and q_{contact} are treated as behavioural scenario parameters rather than fitted coefficients. The values $p_{\text{int}} = 0.30 \text{ h}^{-1}$ and $q_{\text{contact}} = 0.20 \text{ h}^{-1}$ are set so that broadcast dominates peer contact, consistent with AECOM (2017); these parameters govern the timing of awareness, not the topology of feasible paths, and should be interpreted as stress-test assumptions rather than calibrated behavioural estimates. An initial informed share of 0.05 is a conservative assumption for sudden-onset disruption.⁵

Disruption response. The probability that an informed agent opts not to travel, $p_{\text{reschedule}} = 0.13$: consistent with the 13% ‘choose not to travel’ rate observed among non-London long-distance rail passengers facing service disruption (AECOM, 2017, Table 3.4). Rerouting interval $r = 4 \text{ h}$: a modelling assumption representing a realistic re-planning cycle for an informed agent mid-journey; in full-node-closure scenarios the daytime service window overrides this interval, making stranding geography analytically insensitive to its exact value.

Event-level plausibility check against Storm Ciaran. The stranding geography produced by the network and demand layers is compared against Storm Ciaran (1–2 November 2023) as an event-level plausibility check. The information parameters (p_{int} , q_{contact}) are not validated here: they govern awareness timing, not which cities strand, and stranding is topology-determined. OPDI disruption severity per city (fractional flight reduction vs. a same-weekday baseline) correlates positively with ABM-simulated stranding counts across 34 cities ($\rho = 0.498$, $p = 0.003$; Figure 3), supporting external plausibility. This should be interpreted as an event-level plausibility check rather than full predictive validation, as the same OPDI dataset underlies both air network construction and the severity measure.

⁵D’Arienzo et al. (2020) and O’Shea et al. (2020) motivate the two-channel warning structure but are not used numerically, as their empirical settings differ from disrupted transport. At $p_{\text{int}} = 0.30 \text{ h}^{-1}$, independent broadcast awareness reaches approximately 66% after three hours and 88% after six hours (discrete per-step formula); the grid upper bound $p_{\text{int}} = 0.50 \text{ h}^{-1}$ implies approximately 98%, representing a near-instantaneous scenario. AECOM (2017) reports high pre-travel awareness for *planned* disruptions (74–76%) and is used here only as a behavioural reference for channel ranking and rescheduling response, not as direct calibration for sudden-onset awareness.

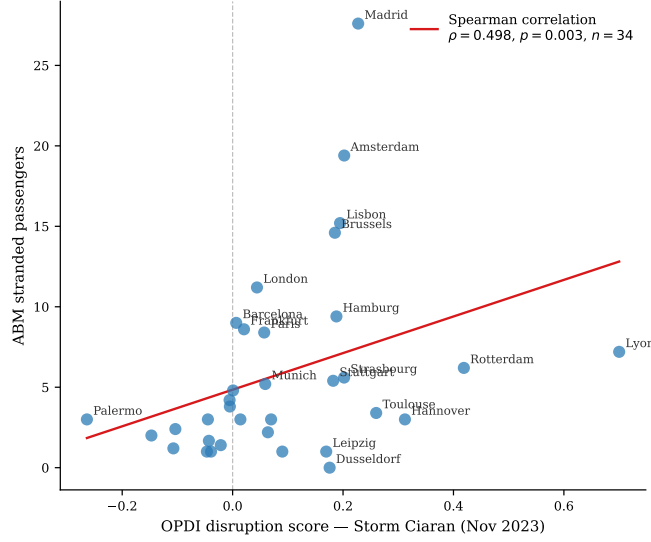


Figure 3: Storm Ciaran event-level plausibility check ($\rho = 0.498$, $p = 0.003$, $n = 34$). x -axis: OPDI disruption severity (fractional flight reduction vs. baseline). y -axis: ABM-simulated mean stranded agents.

5 Results

5.1 Baseline Connectivity Check

Zero agents strand across all five Monte Carlo baseline runs ($\sigma_c = 0$ for all cities), confirming that all sampled OD pairs remain routable under normal service.

5.2 Scenario Outcomes

Scenario A: Northern Storm. Closing London, Amsterdam, and Hamburg produces the highest total stranding burden across all three scenarios. The most striking pattern is *secondary stranding*: hinterland cities (Essen, Dortmund) outrank closed hubs because they depend on Amsterdam or Hamburg for nearly all intercity connections, illustrating the interdependency regime of Buldyrev et al. (2010). Stranding stabilises within hours of onset: information diffusion affects timing but not the final stranded count, as the binding constraint is topological.

Scenario B: Alpine Freeze. Total stranded passengers are substantially lower than in Scenario A because air alternatives from Milan, Vienna, and Geneva remain operational. Zurich is the sole focus of stranding (Table 3); the second-highest city, Palermo, has a stranding index almost seven times lower, illustrating the focused nature of the Alpine disruption. This produces the engineering/ecological resilience contrast of Reggiani et al. (2015): Zurich exhibits hard engineering failure (stranding), while neighbouring cities show elevated delay only. Informed agents successfully reroute around partial Alpine blockages, suggesting that awareness improves outcomes where alternatives exist (Leng and Corman, 2020).

Scenario C: Atlantic Cascade. The Atlantic Cascade produces the broadest secondary stranding footprint (Table 3). Travellers associated with Bilbao and Murcia have the highest stranding exposure despite those cities not being directly closed (Table 3): both depend entirely on rail-air corridors through Madrid or Iberian coastal routes severed by the cascade. Bilbao’s index should be interpreted cautiously: it has no air OD pairs in the gravity model due to OPDI transponder coverage gaps (Section 6), so its isolation partly reflects absent air alternatives in the data rather than the city’s true connectivity. Lisbon’s position is critical: with no direct long-distance rail alternative to Spain in the modelled network, node closure severs every access path. Madrid’s

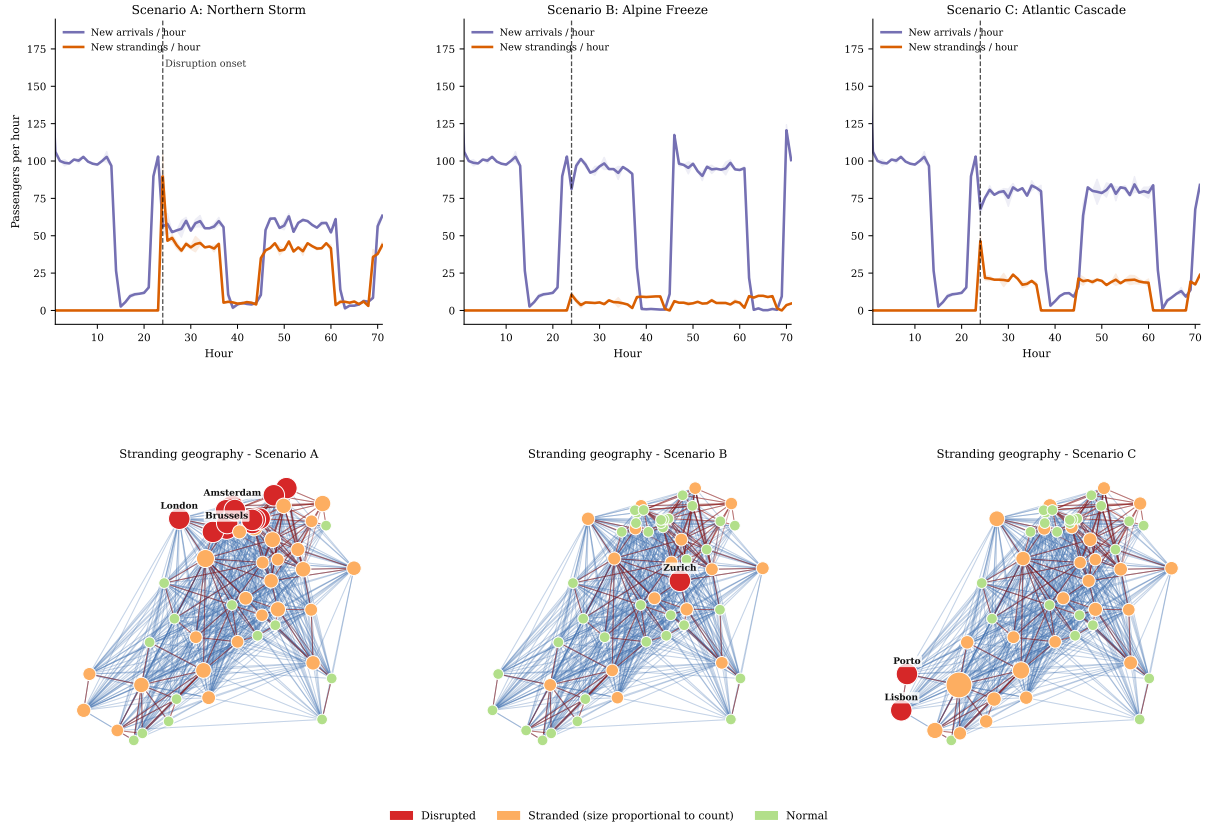
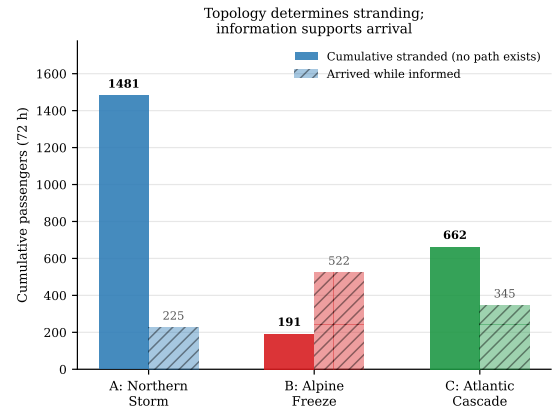


Figure 4: *Top:* per-hour arrivals and strandings (mean $\pm$ SD, 5 runs); dashed line at hour 24 marks disruption onset; overnight dips reflect service-window closure (23:00–05:00). *Bottom:* stranding geography.

Table 3: Top origin/destination exposure cities by stranding index. Hinterland cities without direct closure consistently outrank closed hubs.

City	σ_c	Raw count	Type
<i>Scenario A: Northern Storm</i>			
Essen	0.339	52	Hinterland
Brussels	0.332	98	Hinterland
Bremen	0.332	33	Hinterland
London	0.331	235	Closed hub
Dortmund	0.330	61	Hinterland
Hamburg	0.327	94	Closed hub
Amsterdam	0.322	142	Closed hub
<i>Scenario B: Alpine Freeze</i>			
Zurich	0.319	92	Closed hub
Palermo	0.047	2	Hinterland
Strasbourg	0.036	6	Hinterland
<i>Scenario C: Atlantic Cascade</i>			
Bilbao	0.364	12	Hinterland
Murcia	0.357	11	Hinterland
Madrid	0.332	175	Hinterland
Seville	0.329	39	Hinterland
Porto	0.325	40	Closed hub
Lisbon	0.322	85	Closed hub

Figure 5: Cumulative stranded vs. arrived-while-informed over 72 h by scenario. In Scenario B informed arrivals exceed strandings; in A and C stranding dominates, supporting the interpretation that awareness helps mainly where alternatives exist.



disrupted air corridors propagate isolation across the south; the spatial compounding (Zscheischler et al., 2020) is evident across the geographic arc from Lisbon to Paris.

5.3 Information-Layer Dynamics

Figure 5 supports the interpretation that information improves ecological resilience (Scenario B, where alternatives exist) but cannot create engineering resilience where none exists (Cats and Jenelius, 2014; Leng and Corman, 2020; Reggiani et al., 2015).

Table 4: Information-pathway summary per scenario (mean across 5 MC runs, step 72). “External” denotes broadcast/app/official information; “Peer” denotes mean-field contact transmission; “Experiential” denotes blocked-edge or forced discovery after disruption exposure.

Scenario	External %	Peer %	Experiential %	Arrived-informed	Stranded
A: Northern Storm	17.3	0.5	82.1	225	1481
B: Alpine Freeze	54.9	6.1	39.0	522	191
C: Atlantic Cascade	35.1	3.1	61.8	345	662

Peer contact is a minor channel; broadcast dominates in Scenario B, where partial disruption extends the proactive window (Leng and Corman, 2020). In Scenarios A and C, experiential discovery accounts for over 60% of awareness events, reflecting the sudden compound onset that leaves little proactive broadcast window.

5.4 Centrality and Stranding Correlation

Betweenness centrality shows moderate positive correlation with stranding in Scenario A ($r = 0.418$), where the disrupted nodes are themselves the highest-centrality cities. The relationship is weak in Scenarios B ($r = 0.048$) and C ($r = 0.185$), where stranding is driven by geographic proximity to the disruption arc rather than network position. Direct exposure therefore dominates hard stranding; centrality amplifies secondary rerouting burden only when the disrupted nodes are major hubs.

5.5 Sensitivity and Parametric Robustness

Spawn-rate sensitivity ($\lambda \in \{100\text{--}1000\} \text{ h}^{-1}$) confirms stranding rates σ_c are stable across a tenfold agent-population range. Disruption probability sensitivity ($p_{\text{disrupt}} \in [0.5, 1.0]$) preserves city rank ordering; Scenario C shows a threshold effect near full closure where the last Iberian paths fail simultaneously. Air-overhead sensitivity (45, 90, and 120 min boarding assumptions) produces identical top-stranded city rankings across all scenarios and changes total stranding by less than 3% (Scenario C: 662 to 678); shortest-path routing and modal split are robust to the overhead parameterisation. Information-parameter sensitivity across 16 combinations ($p_{\text{int}} \in \{0.10, 0.20, 0.30, 0.50\} \text{ h}^{-1}$, $q_{\text{contact}} \in \{0.05, 0.10, 0.20, 0.40\} \text{ h}^{-1}$) spans a range from slow diffusion to near-instantaneous broadcast awareness (the upper bound $p_{\text{int}} = 0.50 \text{ h}^{-1}$ implies approximately 98% independent awareness within six hours): Scenario B arrived-informed exceeds A and C in all 16 combinations, stranding deviates by at most 1% (A) and 2% (C), and top-ranked cities are identical across the full grid, confirming that topology dominates awareness in Scenarios A and C.

Compound ablation. To isolate the interdependency contribution, each scenario was re-run under rail-only and air-only disruption with scenario geometry held constant (Table 5). In Scenarios B and C neither single mode alone produces stranding: compound disruption uniquely removes both modal alternatives simultaneously. In Scenario A, rail-only disruption produces

just 21% of the compound total, confirming that simultaneous air and rail closure drives most stranding. These results align with the Buldyrev et al. (2010) interdependency mechanism.

Table 5: Compound ablation: total stranded agents by disruption mode.

Scenario	Compound	Rail-only	Air-only
A: Northern Storm	1,481	310	0
B: Alpine Freeze	191	0	0
C: Atlantic Cascade	662	0	0

6 Conclusions

This paper simulated passenger redistribution across a 53-city European multimodal transport network under three compound weather-disruption scenarios, compared against Storm Ciaran as an external plausibility check. Three main findings emerge.

First, *topology is the binding constraint on hard stranding*. Direct node closures produce stranding that information diffusion cannot overcome: when no path exists, awareness of that fact does not create one. A compound ablation supports the interdependency mechanism described by Buldyrev et al. (2010): single-mode disruption alone produces little or no stranding across scenarios, while compound disruption produces substantially more, underlining that simultaneous multi-modal failure is qualitatively distinct from either mode failing in isolation.

Second, *peripheral air-dependent cities are the most vulnerable*. Lisbon, with no direct long-distance rail alternative to Spain in the modelled network, becomes a topological island under node closure. Bilbao and Murcia, though not directly closed, record the highest stranding indices in Scenario C (Table 3) because all their connections route through disrupted corridors; Bilbao’s exact value should be interpreted cautiously given missing air OD coverage (see Limitations), but Murcia’s ranking is unaffected by that gap. The results suggest that compound weather risk is not uniformly distributed but concentrates at cities with structurally limited multimodal alternatives, a pattern with direct implications for network resilience planning by bodies such as ERA and EUROCONTROL.

Third, *information timing matters only where alternatives exist*. The awareness layer has measurable effects in Scenario B, where partial connectivity survives, but negligible effects in Scenarios A and C. Information improves ecological resilience only where alternatives survive; where topology removes all paths, awareness cannot prevent hard stranding (Reggiani et al., 2015); this result is invariant to broadcast and peer-contact rate variation (Section 5).

Limitations. The model captures topological isolation under static disruptions, not congestion on surviving routes or recovery dynamics; each city is a single node and demand is gravity-modelled rather than from observed origin-destination matrices. OPDI covers only actual ADS-B departures, so delay-dominated events produce attenuated validation signals, and coverage gaps leave Bilbao and three other cities with missing air OD pairs, so Bilbao’s stranding index partly reflects absent data rather than pure network topology. Sensitivity sweeps for p_{init} and $p_{\text{reschedule}}$ were not performed; as these affect awareness timing rather than path existence, they are unlikely to alter city rankings, but this remains a limitation.

Future work. Priority extensions include temporal recovery dynamics, capacity-constrained rerouting on surviving corridors, and probabilistic disruption-intensity sweeps. Pre-planned agent types (departures hours away when the warning arrives) would isolate the causal effect of early information on stranding outcomes.

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